FLAT GRADED ALGEBRAS AND FLAT FINITE GRADED MODULES ARE FINITELY PRESENTED OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- A flat finite graded module over a graded polynomial ring over a local ring is finitely presented. Lemma \ref{lemma-flat-graded-finite-type-finite-presentation-module}, cited at proof lines 5936-5938. Gives finite presentation after localising at each of the primes.
- Local finite presentation plus injectivity into the product of the localisations gives global finite presentation. Lemma \ref{lemma-flat-finite-type-finite-presentation-local-module}, cited at proof lines 5939-5940. Glues the local statements into the required global one.

Source reference. Original lemma 053D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6601–6601. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 053C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6600–6600. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 053A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6598–6598. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring. Let $S = \bigoplus_{n \geq 0} S_n$ be a graded R-algebra. Let $M = \bigoplus_{d \in \mathbf{Z}} M_d$ be a graded S-module. Assume S is finitely generated as an R-algebra, assume S_0 is a finite R-algebra, and assume there exist finitely many primes $\mathfrak{p}_j$, $i = 1, \ldots, m$ such that $R \to \prod R_{\mathfrak{p}_j}$ is injective.*

(1) *If S is flat over R, then S is a finitely presented R-algebra.*
(2) *If M is flat as an R-module and finite as an S-module, then M is finitely presented as an S-module.*

Proof. As S is finitely generated as an R-algebra, it is finitely generated as an S_0 algebra, say by homogeneous elements $t_1, \ldots, t_n \in S$ of degrees $d_1, \ldots, d_n > 0$. Set $P = R[x_1, \ldots, x_n]$ with $\deg(x_i) = d_i$. The ring map $P \to S$, $x_i \to t_i$ is finite as S_0 is a finite R-module. To prove (1) it suffices to prove that S is a finitely presented P-module. To prove (2) it suffices to prove that M is a finitely presented P-module. Thus it suffices to prove that if $S = P$ is a graded polynomial ring and M is a finite S-module flat over R, then M is finitely presented as an S-module. By Lemma 053C we see $M_{\mathfrak{p}}$ is a finitely presented $S_{\mathfrak{p}}$-module for every prime $\mathfrak{p}$ of R. Thus the result follows from Lemma 053A. □